

Driving Travel Time Comparison and Analysis

The aim of this project is to analyze the factors that affects my daily car travel time and to understand how different conditions influence the trip duration. The project will depend on real-life driving patterns and aims to explore whether travel time can be explained and predicted using simple variables such as traffic level (1 to 3), time of day, and distance

The data will be collected between April 1 and April 14 from regular car trips made by me, my friends, and my family. For each trip, I will collect information including departure time, arrival time, number of traffic lights and actual travel duration. In addition, I will also record distance and estimated travel time from Google Maps, traffic level, and whether the trip occurs in rush hours like after work.

It is expected that I will collect approximately 6 to 10 trips per day, resulting in an estimated dataset of 100 to 150 observations. The data will be organized in a structured format where each row represents a trip and each column corresponds to a specific variable. This dataset will be sufficient for performing exploratory data analysis, statistical testing, and basic machine learning modeling.

The analysis will examine how travel time changes with respect to traffic conditions, time of departure, and distance. Additionally, the difference between estimated and actual travel times will be analyzed to evaluate the accuracy of navigation tools. Finally, a simple predictive model will be built to estimate travel duration based on the collected features.

This project follows the main steps of the data science process, including data collection, data analysis, and modeling, and aims to provide insights into personal driving efficiency and urban traffic behavior.